

Midterm Exam (Version 1)

CAS CS 132: Geometric Algorithms

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- ▷ You will have approximately 75 minutes to complete this exam. Make sure to read every question, some are easier than others.
- ▷ Do not remove any pages from the exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.
- ▷ We will not look at any work on the pages marked "*This page is intentionally left blank.*" You should use these pages for scratch work.

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1 Matrix Equations

$$\begin{matrix} \text{A} & & \text{B} \\ \begin{bmatrix} 3 & -2 & -6 & 1 \\ 5 & -3 & -3 & 1 \\ 5 & 2 & 1 & 2 \end{bmatrix} & \mathbf{x} = & \begin{bmatrix} 1 & 1 & 2 & 0 \\ 4 & -1 & 3 & 0 \\ 7 & 1 & 1 & 1 \end{bmatrix} \mathbf{x} \end{matrix}$$

(10 points) Determine a general form solution for the above equation. That is, determine a general form solution that describes all vectors in $\mathbb{R}^4$ that satisfy the above equality.

$$A - B = \begin{bmatrix} 2 & -3 & -8 & 1 \\ 1 & -2 & -6 & 1 \\ -2 & 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -6 & 1 \\ 2^{-2} & -3^{+4} & -8^{+12} & 1^{-2} \\ -2^{+2} & 1^{-4} & 0^{-12} & 1^{+2} \end{bmatrix} \sim \begin{bmatrix} 1 & -2^{+2} & -6^{+8} & 1^{-2} \\ 0 & 1 & 4 & -1 \\ 0 & -3 & -12 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

$$x_1 = -2x_3 + x_4$$

$$x_2 = -4x_3 + x_4$$

x_3 is free

x_4 is free

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2 True/False

(8 points) Determine if each of the following statements is **True** or **False**. Bubble in your answers below.

- A. (1 point) For any matrix $A \in \mathbb{R}^{m \times n}$, if $m > n$ then the columns of A are linearly dependent.
- ☐ True
☒ False
- B. (1 point) If A and B are row equivalent and U is an echelon form of A then U is also an echelon form of B .
- ☒ True
☐ False
- C. (1 point) A *consistent* linear system whose coefficient matrix is square and has linearly independent columns has infinitely many solutions.
- ☐ True
☒ False
- D. (1 point) If ~~A is~~ the augmented matrix of a linear system ~~and A~~ has a pivot in every column then the linear system is inconsistent.
- ☒ True
☐ False
- E. (1 point) If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent then so is $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_3\}$.
- ☒ True
☐ False
- F. (1 point) For $A \in \mathbb{R}^{m \times n}$, if the columns of A span $\mathbb{R}^m$, then the columns of A^T are linearly independent.
- ☒ True
☐ False
- G. (1 point) The matrix that implements a linear transformation is unique.
- ☒ True
☐ False
- H. (1 point) $AB \neq BA$ for all matrices $A, B \in \mathbb{R}^{n \times n}$.
- ☐ True
☒ False

3 Stochastic Matrices

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 0 & 1 \\ 3 & 5 & 1 \end{bmatrix}$$

A. (2 points) Calculate $\mathbf{1}^T A$ where $\mathbf{1}$ is the all-ones vector in $\mathbb{R}^3$.

$$\begin{array}{ccc} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 4 & 1 \\ 2 & 0 & 1 \\ 3 & 5 & 1 \end{bmatrix} & = \begin{bmatrix} (1+2+3) & (4+5) & (1+1+1) \end{bmatrix} \\ 1 \times 3 & 3 \times 3 & \\ & & = \begin{bmatrix} 6 & 9 & 3 \end{bmatrix} \\ & & 1 \times 3 \end{array}$$

Solution.

$$\begin{bmatrix} 6 & 9 & 3 \end{bmatrix}$$

B. (4 points) A matrix B is **stochastic** if its entries are non-negative and $B^T \mathbf{1} = \mathbf{1}$ (this implies the sum of the entries in each column of B is 1). Determine a diagonal matrix C such that AC is stochastic.

$$(AC)^T \mathbf{1} = C^T A^T \mathbf{1} = C (\mathbf{1}^T A)^T = C \begin{bmatrix} 6 & 9 & 3 \end{bmatrix}^T = \mathbf{1}$$

$$\begin{bmatrix} c_1 & 0 & 0 \\ 0 & c_2 & 0 \\ 0 & 0 & c_3 \end{bmatrix} \begin{bmatrix} 6 \\ 9 \\ 3 \end{bmatrix} = \begin{bmatrix} 6c_1 \\ 9c_2 \\ 3c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \begin{array}{l} c_1 = \frac{1}{6} \\ c_2 = \frac{1}{9} \\ c_3 = \frac{1}{3} \end{array}$$

Solution.

$$\begin{bmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$$

C. The matrix A is reproduced for convenience.

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 0 & 1 \\ 3 & 5 & 1 \end{bmatrix}$$

(4 points) Let C be the matrix you determined in the previous part. Calculate $\mathbf{1}^T A C [-1 \ 5 \ 3]^T$.

$$\begin{aligned} \mathbf{1}^T A C [-1 \ 5 \ 3]^T &= ((A C)^T \mathbf{1})^T [-1 \ 5 \ 3]^T \\ &= \mathbf{1}^T [-1 \ 5 \ 3]^T \\ &= [1 \ 1 \ 1] \begin{bmatrix} -1 \\ 5 \\ 3 \end{bmatrix} \\ &= -1 + 5 + 3 \end{aligned}$$

Solution.

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4 Increasing Span

$$[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \mathbf{a}_4 \ \mathbf{a}_5 \ \mathbf{a}_6] \sim \begin{bmatrix} 1 & -3 & -11 & -2 & 9 & 1 \\ 0 & -2 & -8 & -4 & 6 & -5 \\ 0 & 0 & 4 & 4 & -2 & 1 \\ 0 & 0 & 0 & 0 & 8 & 5 \end{bmatrix}$$

(10 points) Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5, \mathbf{a}_6$ be vectors such that the above row equivalence holds. Determine the *first* vector in this list of vectors which lies in the span of the previous vectors in the list. That is, determine the vectors $\mathbf{a}_i$ with the smallest index i such that $\mathbf{a}_i \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_{i-1}\}$. Furthermore, write $\mathbf{a}_i$ as a linear combination of the vectors $\mathbf{a}_1, \dots, \mathbf{a}_{i-1}$.

$$\begin{bmatrix} 1 & -3 & -11 & -2 \\ 0 & -2 & -8 & -4 \\ 0 & 0 & 4 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -11 & -2 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 0 & 9 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Note: We dropped the last row for convenience

Solution.

$$\vec{v}_4 = 3\vec{v}_1 - 2\vec{v}_2 + \vec{v}_3$$

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5 Linear Transformations

A.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 2x_2 - 3x_1 \\ 0 \\ x_2 + 4x_1 \end{bmatrix}$$

(3 points) Let $S : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear transformation above. Determine the matrix that implements S . That is, determine the matrix A such that $S(\mathbf{v}) = A\mathbf{v}$ for all vectors $\mathbf{v} \in \mathbb{R}^2$.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} -3 \\ 0 \\ 4 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Solution.

$$\begin{bmatrix} -3 & 2 \\ 0 & 0 \\ 4 & 1 \end{bmatrix}$$

B.

$$x + 3y + z = 0$$

(3 points) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation that projects vectors vertically onto the point-set of the above linear equation. For example,

$$T\left(\begin{bmatrix} 2 \\ -4 \\ 9 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -4 \\ 10 \end{bmatrix}$$

because $2 + 3(-4) + 10 = 0$. Determine the matrix that implements T .

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solution.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -3 & 0 \end{bmatrix}$$

- C. (4 points) Let S and T be the linear transformations from the previous two parts. Determine a **nonzero** vector that is in both the range of S and the range of T (i.e., determine a vector in $\text{ran}(S) \cap \text{ran}(T)$). Justify your answer.

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\text{ran}(S) = \left\{ \begin{bmatrix} x \\ 0 \\ z \end{bmatrix} : x, z \in \mathbb{R} \right\}$$

The columns of the matrix that implements S are L.I. and both lie in the plane $y = 0$, so must span the whole plane.

Solution.

$$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Justification.

$$\text{ran}(S) = \left\{ \begin{bmatrix} x \\ 0 \\ z \end{bmatrix} : x, z \in \mathbb{R} \right\} \text{ and } T(\vec{e}_1) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \in \text{ran}(S)$$

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